

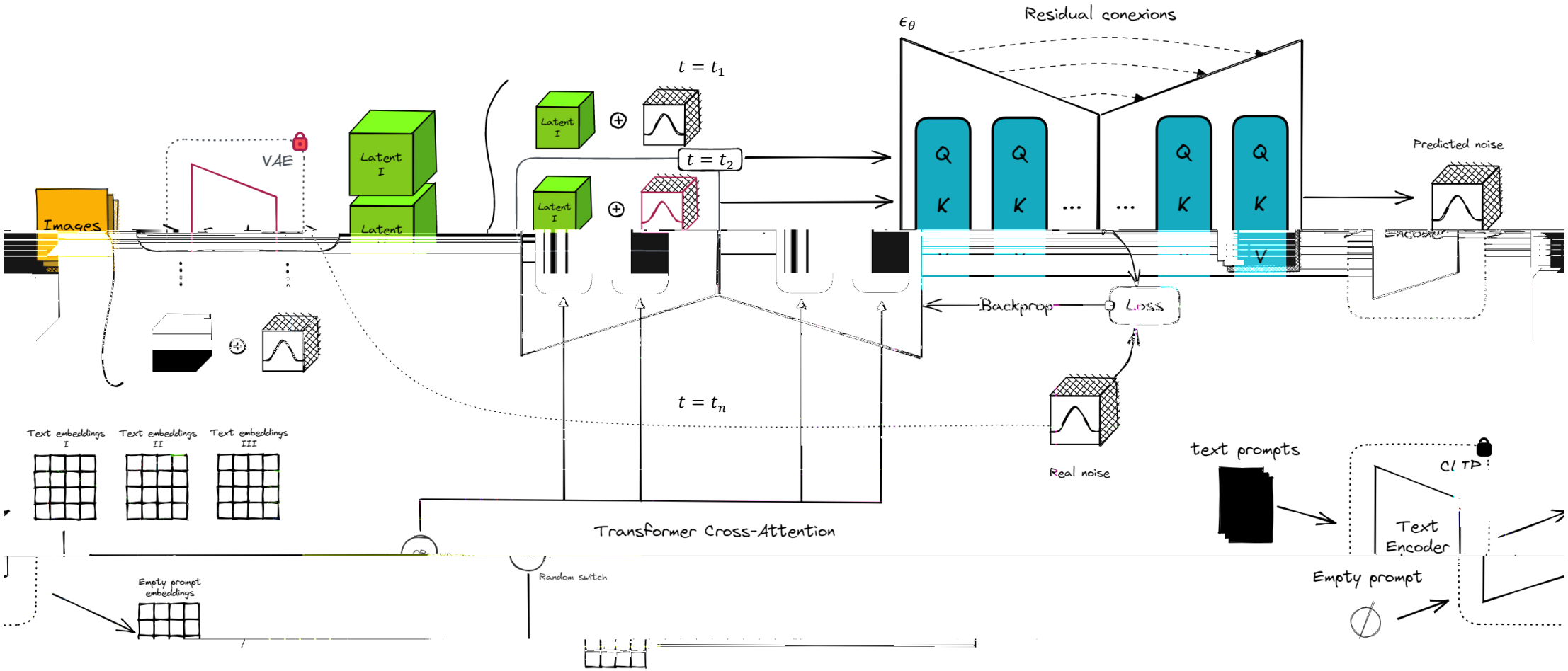
Expandiendo Stable Diffusion

N

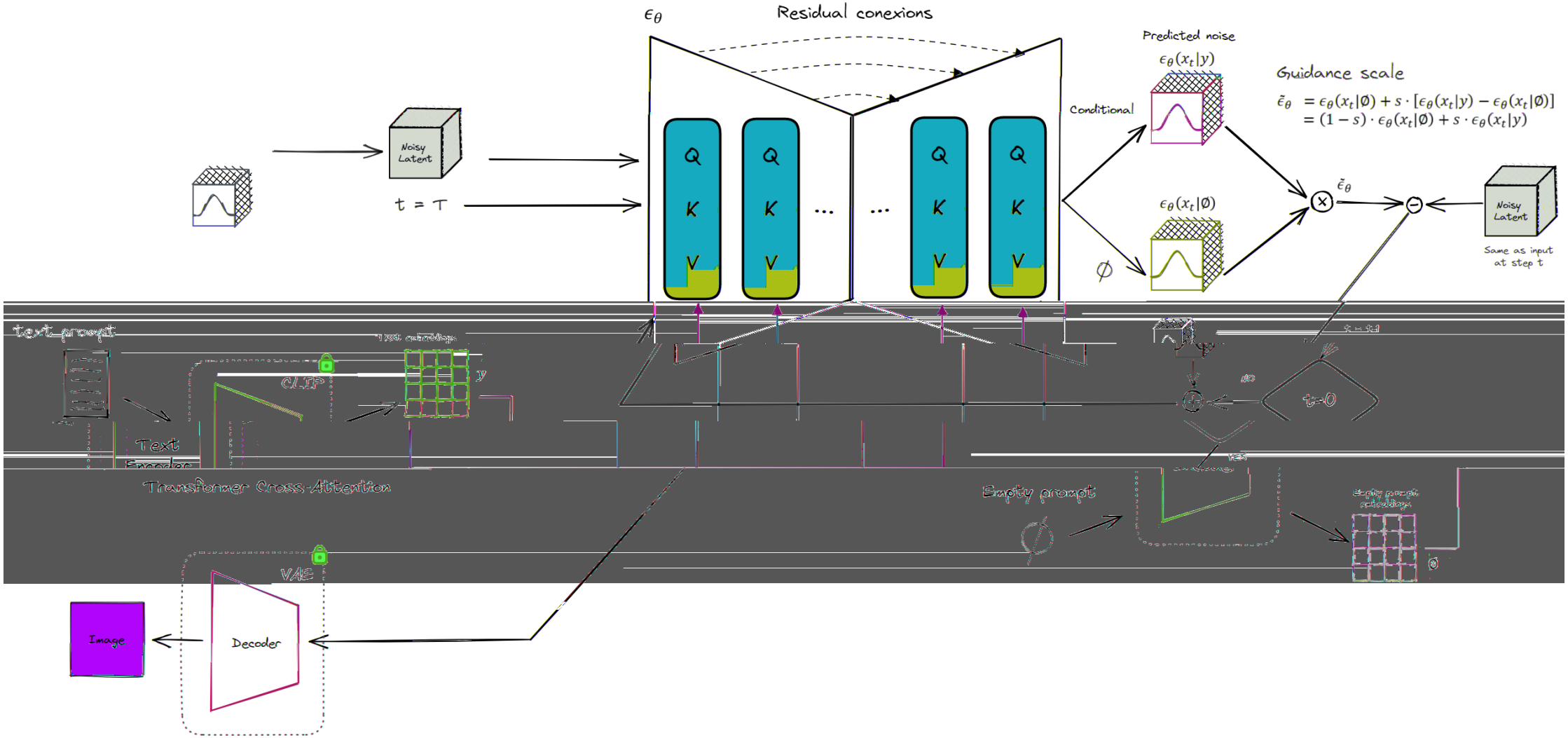
Adrian Mikel Maiza Coupin

upna
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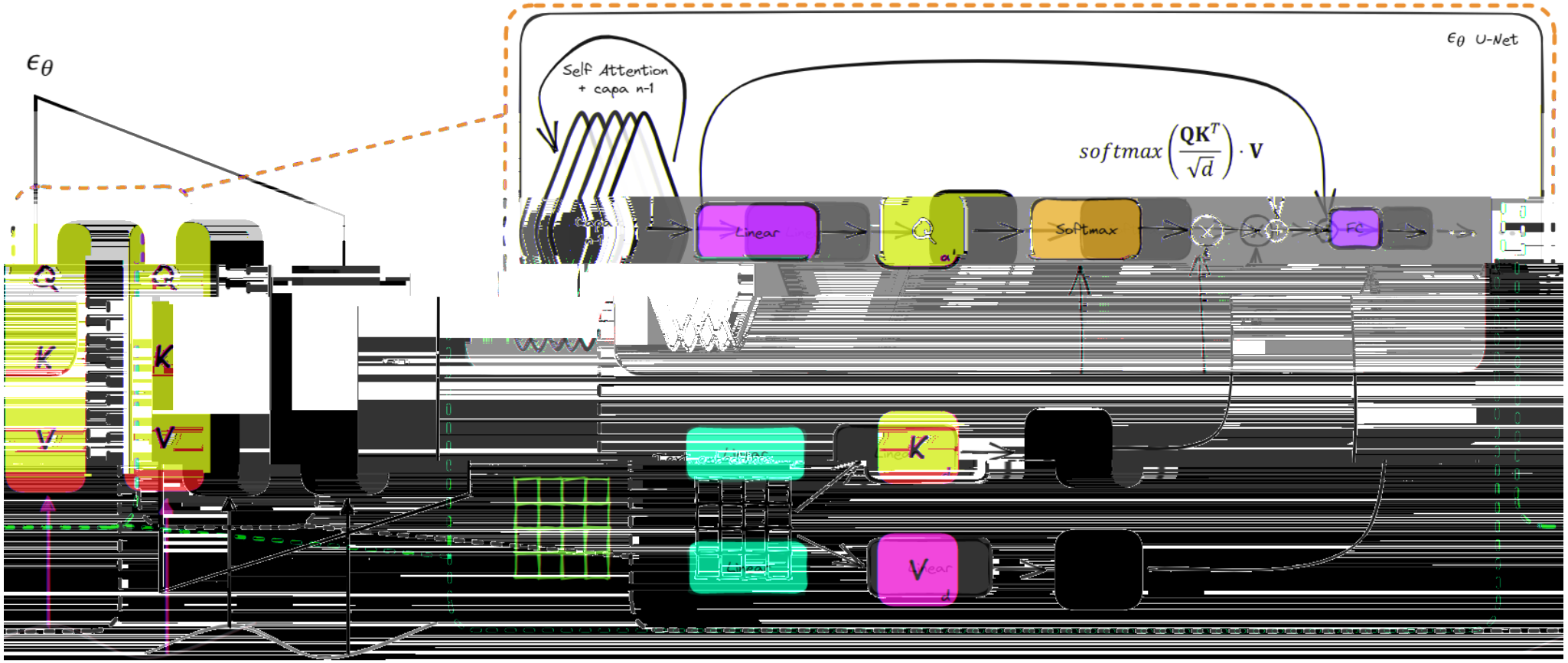
01 Recap: Entrenamiento SD



01 Recap: Inferencia SD

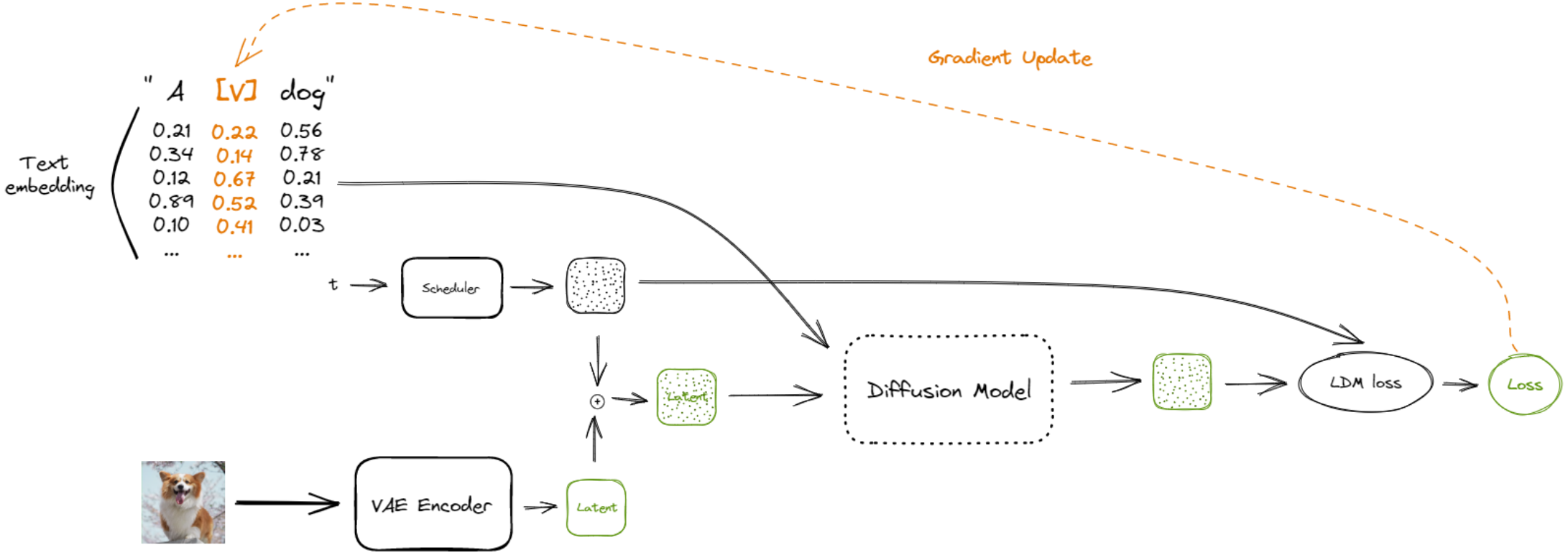


01 Recap: Inferencia SD



Objetivos

02 Textual Inversion



02 Dreambooth

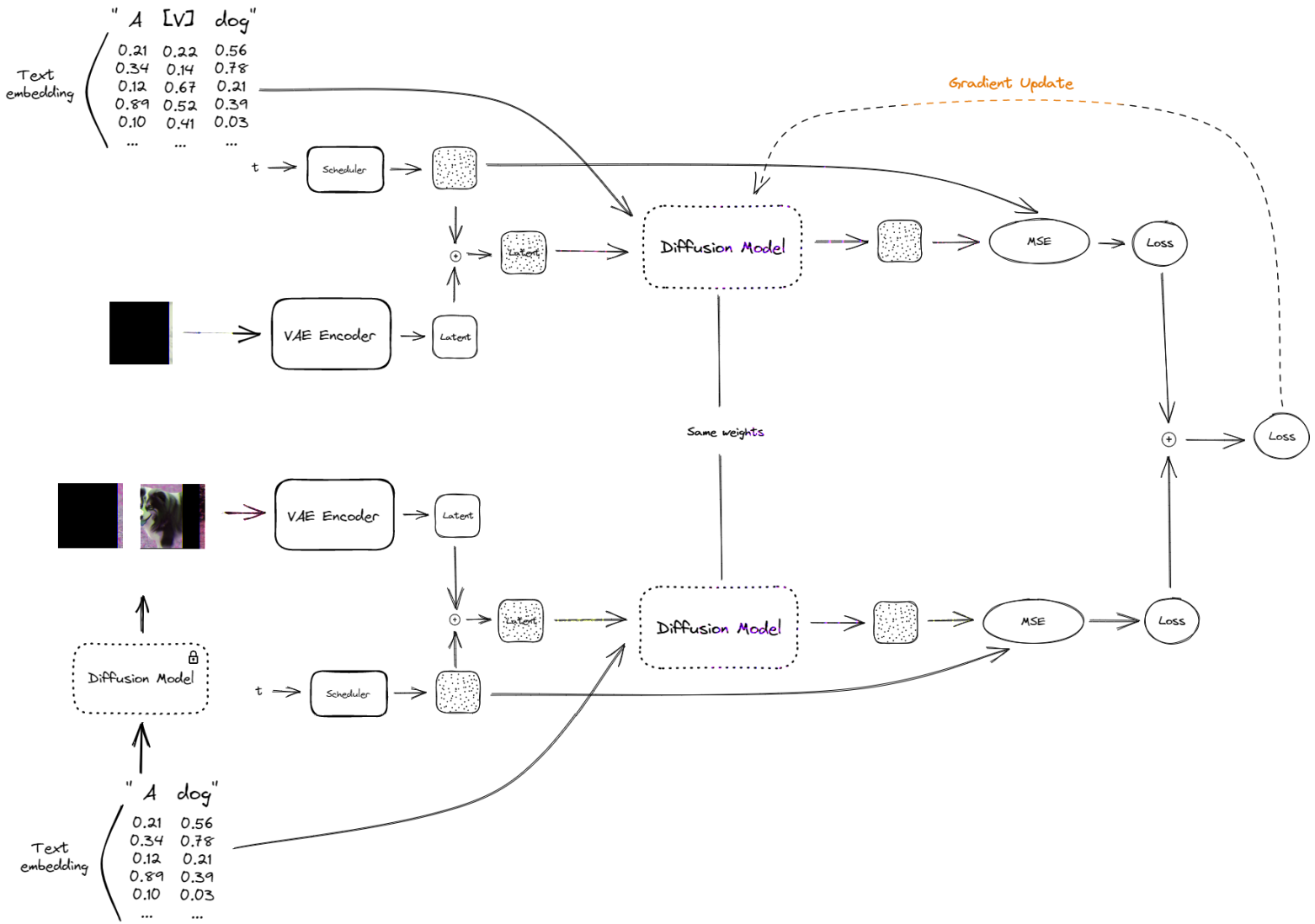
"A [identifier] [class noun]"

1) Utilizar una palabra existente como identificador (único/especial) ->
   Aumenta el tiempo de entrenamiento y disminuye el rendimiento

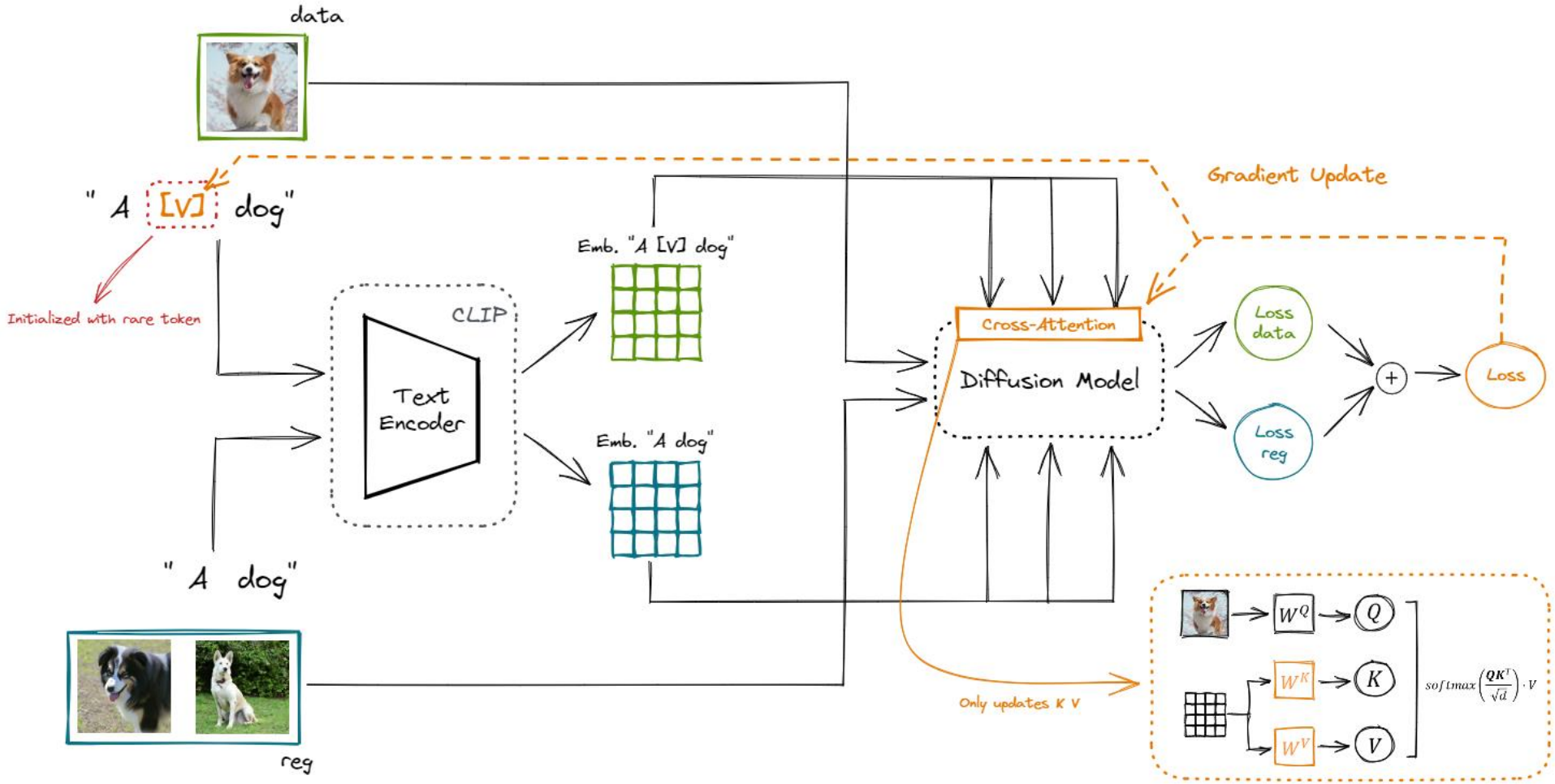
2) Generar un identificador raro mezclando letras/números aleatorios (xxy5st) ->
   El tokenizador podría tokenizar cada letra por separado

3) Encontrar un token raro en el vocabulario y luego invertirlos en el espacio de texto
   Al final se buscan palabras con longitud 3

02 Dreambooth



02 Custom Diffusion



$V^* \in P_{200k}$, donde $P_{200k} \subseteq \text{LAION}_{400\text{M}}$

V^* c $5 \leq count(V^*) \leq 10$ V^* c c

02 LoRA

finetuning

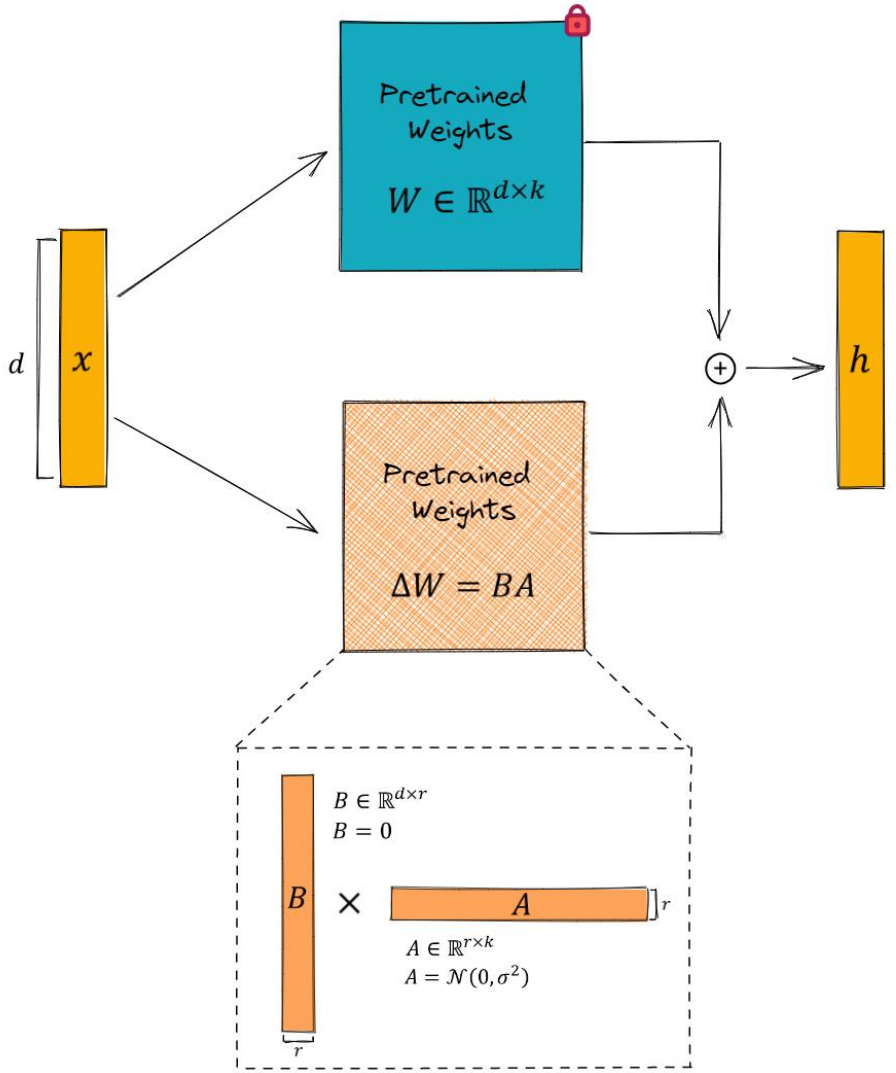
"Large Language Models" (N

c

ΔW c B A r

C

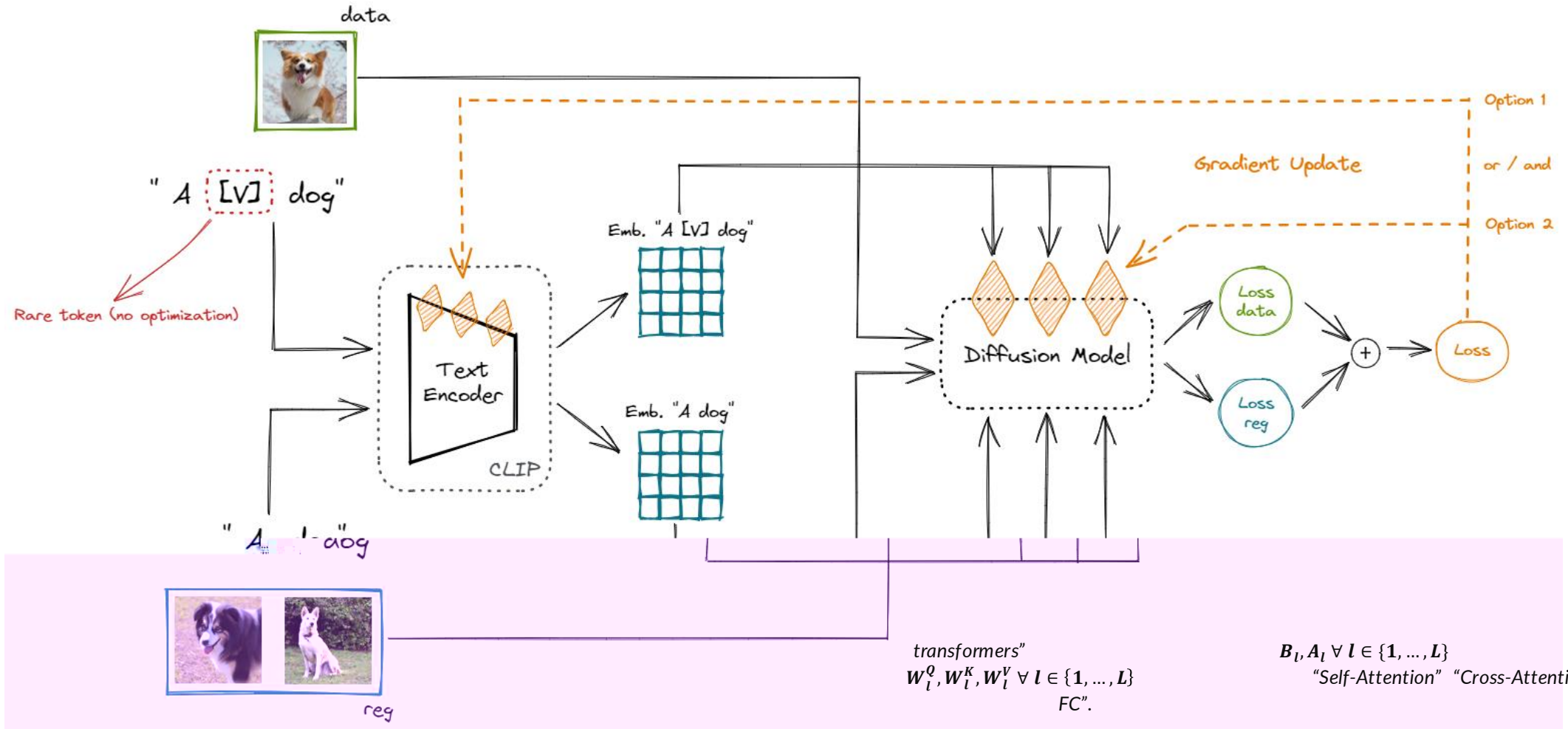
A



$$h = W'x = Wx + \Delta Wx = Wx + BAx$$

$$W \in \mathbb{R}^{d \times k}, B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}, r \ll \min(d, k)$$

02 Dreambooth con LoRA



transformers"
$W_l^Q, W_l^K, W_l^V \; \forall \; l \in \{1, \ldots, L\}$
FC".

$B_l, A_l \; \forall \; l \in \{1, \ldots, L\}$
"Self-Attention" "Cross-Attention"